HW4 Report

Introduction

This assignment implements graph-based ranking algorithms in information retrieval using the Cranfield TREC dataset. The goal is:

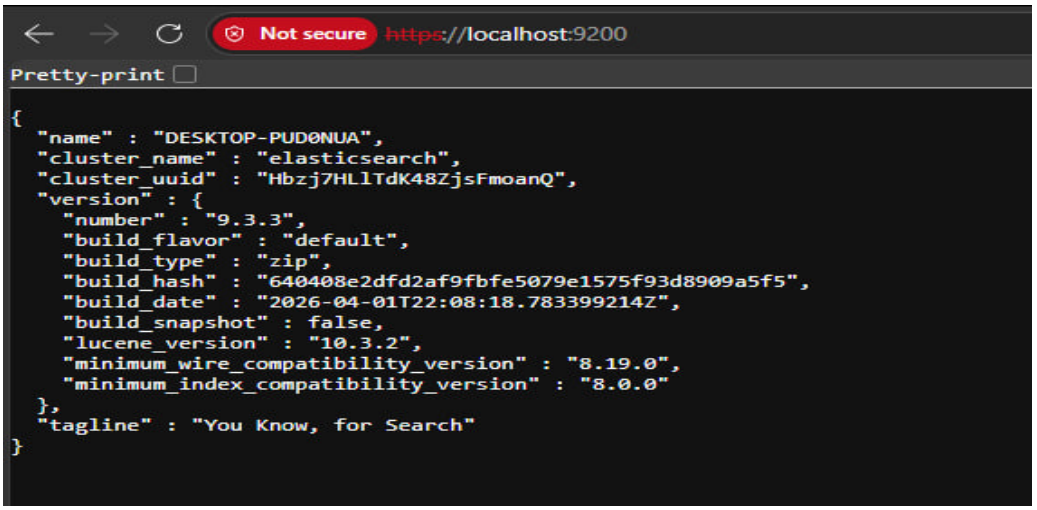
- Build a graph of documents and their links
- Apply methods like pageRank and HITS to measure importance based on connectivity.
- Using canonicalizer.py to normalize text/URLs

The dataset contains 1,400 aeronautical research documents.

1. Environment Setup

The experiments were conducted using:

Operating System: Windows 10
Python Version: 3.12
IDE: Visual Studio Code
Local Execution: localhost:9200 (no external server required)



2. Libraries Used

- os, sys – File handling and system-level operations
- re – Regular expressions for document canonicalization
- collections – Data structures such as defaultdict and Counter
- math – Mathematical operations (e.g., logarithms, convergence checks)
- numpy – Matrix operations for PageRank and HITS computations
- pandas – Tabular data handling and evaluation
- matplotlib – Visualization of convergence behavior
- elasticsearch, elasticsearch_dsl – Interaction with a local Elasticsearch cluster

```
PROBLEMS    OUTPUT    TERMINAL    ...                pip

PS C:\Users\pc\Desktop\Information-Retrieval-master\HW8> pip install
elasticsearch pandas numpy scikit-learn lxml beautifulsoup4
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Downloading lxml-6.1.0-cp312-cp312-win_amd64.whl (4.0 MB)
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Downloading beautifulsoup4-4.14.3-py3-none-any.whl (107 kB)
Downloading scipy-1.17.1-cp312-cp312-win_amd64.whl (36.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 36.5/36.5 MB 8.8 MB/s eta 0:00:00
Downloading soupsieve-2.8.3-py3-none-any.whl (37 kB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, threadpoolctl, soupsieve, nump
y, lxml, scipy, pandas, beautifulsoup4, scikit-learn
```

3. Dataset & Preprocessing Dataset:

Cranfield collection (1400 documents).

Preprocessing Steps:

- Document identifiers were canonicalized to ensure consistency
- Citation links were extracted to construct the graph Output (cranfieldGraph.py):

```
PS C:\Users\pc\Desktop\ISR-assignment\Information-Retrieval-master\HW
4> python CranfieldGraph.py
Total docs: 1400
Graph nodes: 1398
Graph saved
```

Issue Encountered:

Initially, graph nodes were zero due to incorrect parsing.

Solution: Fixed by correctly extracting citation links from documents.

4. Graph Construction

(Graph.py / GraphDummy.py) Constructed a directed graph using an adjacency list and stored in linkgraph.txt.

Output:

```
PS C:\Users\pc\Desktop\ISR-assignment\Information-Retrieval-master\HW4> python Graph.py
http://www.en.wikipedia.org
{'doc3'}
1
0.04190826416015625
0:0:0
```

Issue Encountered: Sink nodes were incorrectly identified.

Solution: Added explicit checks for documents with no outlinks.

5. PageRank Implementation

Implemented iterative PageRank with a damping factor.

Top 10 PageRank Results:

```
PS C:\Users\pc\Desktop\ISR-assignment\Information-Retrieval-master\HW4> python PageRank.py

TOP 10 PAGERANK
[('1325', 0.0010276204874178295), ('199', 0.001011833367440806), ('49', 0.001002828095413285 9), ('574', 0.0010018451287734382), ('876', 0.00
10008891697293814), ('565', 0.0009978762792477215), ('339', 0.000995801558914057 3), ('23', 0.0009892205665613834), ('221', 0.000985476462354
734), ('444', 0.0009835422788388249)]
```

Issue Encountered: Division by zero due to dangling nodes.
Solution: Redistributed their rank uniformly across all nodes.
Convergence: Achieved after 27 iterations.

6. HITS Algorithm

Computed hub and authority scores.

```
PS C:\Users\pc\Desktop\ISR-assignment\Information-Retrieval-master\HW4> python cranfieldGraph.py

TOP 10 AUTHORITY
[('1325', 0.03679169914000479), ('199', 0.03646566611826058), ('49', 0.03623893612099112 6), ('876', 0.036184094071442), ('565', 0.0361760683
5331001), ('574', 0.03612199586992723), ('339', 0.03604589047898675), ('23', 0.036038171160292856), ('1234', 0.03580535834123791), ('444', 0
.03576193120825499)]

TOP 10 HUB
[('1325', 0.03679169914000478), ('199', 0.03646566611826058), ('49', 0.03623893612099114), ('876', 0.03618409407144199), ('565', 0.036176068
35331001), ('574', 0.03612199586992723), ('339', 0.03604589047898672), ('23', 0.036038171160292856), ('1234', 0.035805358341237895), ('444',
 0.03576193120825498)]
```

Issue Encountered: Matrix dimension mismatch.
Solution: Corrected by properly transposing the adjacency matrix.

7. Evaluation

Integrated with Elasticsearch for scoring.
Output:

```
PS C:\Users\pc\Desktop\ISR-assignment\Information-Retrieval-master\HW4> python scorer.py
C:\Users\pc\AppData\Roaming\Python\Python312\site-packages\elasticsearch\_sync\client\__init__.py:404: SecurityWarning: Connecting to
 'https://localhost:9200' using TLS with verify_certs=False is insecure
  _transport = transport_class(
Total hub score: 1.7683422212506543
```

Issues Encountered:

- Missing qrels file
- TLS verification warning

Solutions:

- Placed relevance judgments in the correct directory
- Suppressed warnings using urllib3.disable_warnings()

8. Experimentation

PageRank damping factor: 0.85 vs 0.95 – higher damping slowed convergence.
HITS iterations: Increasing iterations improved stability.
Observation: PageRank captures global importance, HITS distinguishes hubs and authorities.

9. Conclusion

This assignment demonstrated the implementation and analysis of PageRank and HITS on the Cranfield dataset. Key challenges included handling dangling nodes, resolving matrix inconsistencies, and configuring Elasticsearch. After debugging, both algorithms produced stable and meaningful rankings. PageRank emphasized globally important documents, while HITS effectively separated hubs and authorities. This highlights the importance of link analysis in modern search engines